

ART Maritime High Power Very High Frequency (VHF) Emitter System SOW

1.0 Background

The Aircraft Signature and Avionics Measurements (ASAM) Electronic Warfare (EW) group of the Atlantic Ranges and Targets (ART) department operates numerous pulsed radio frequency (RF) mobile emitter systems that can be deployed on land or installed on boats. The mobile emitter systems support research, development, test and evaluation (RDT&E) requirements of military fixed-wing and rotary-wing aircraft as well as Unmanned Aircraft Systems (UAS). The current mobile emitter systems fit into one of two general categories, which are high power or low power. The high power systems meet most customer demands in regard to effective radiated power (ERP) but have limited utility due to low duty cycles and insufficient pulsed waveform capabilities. The low power systems are able to achieve much higher duty cycles and generate complex pulsed waveforms but are limited due to the lower ERP they can produce. To support aircraft RDT&E EW requirements into the future, a maritime high power VHF system that is also capable of handling high duty cycles and producing complex waveforms has been identified as essential.

2.0 Scope

The purpose of this document is to define the key requirements for a new maritime high power VHF emitter system. The requirements are traceable to RDT&E EW user needs.

3.0 System Requirements

The emitter system will be permanently installed onboard a 125 foot boat and will be capable of operating in the open ocean. The design must include an air conditioned enclosure to house all of the equipment in order to provide protection from the elements and cooling while in operation. Additionally, the system is required to have its own generator to supply all electrical power needs during a mission. The design must have a dual antenna system mounted back to back on a pedestal that can rotate in a circular scan. The pedestal should be capable of supporting the weight of the antennas with forces as high as 30g's in all directions. The expectation is that the system will be delivered with manuals and technical drawings and training will be supplied after government acceptance. Spares for any high failure items such as antenna elements and amplifier modules should also be part of submissions. Additional performance requirements are located in Table 1.

Table 1. High Power VHF Emitter Requirements

Name	Requirement
Frequency Range	174 – 220 MHz
Pulse Repetition Frequency (PRF)	Up to 25 kHz
Pulse Width (PW) Range	2 - 2000 us
Modulation on Pulse	Both frequency & phase modulation
Duty Cycle	25%
ERP	100 dBm
Antenna Scan	Circular Scan Adjustable Scan Period from 6 to 30 sec
Antenna Polarization	Horizontal
Antenna Horizontal Beamwidth	Greater Than 6 deg
Signal Source	Programmable across full range of RF, PRF/PRI, PW & pulse modulation types

4.0 Deliverables

One high power VHF emitter system meeting or exceeding the requirements outlined in section 3.0 with associated manuals, drawings and spares.

5.0 Delivery

All items shall be received by the government within one year from contract award.